The Basilisk Quartet: Contract, Script, Blanket, and Ledger as Boundedness Artifacts

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Abstract

We propose a four-artifact security architecture for beneficial general intelligence over a shared typed input/output boundary. A Contract specifies admissible boundary events; a Script realizes events through internal state; a Blanket mediates internal–external dependence; and a Ledger preserves an attested history from which accounts and certificates may be derived. The four artifacts are compatible but non-identical. We prove an initial local robustness theorem, give a finite non-identifiability construction separating Script from Ledger, state the invariant combinatorial form of a Bayesian-network Markov blanket, and introduce future-preserving boundedness to distinguish smooth operation from irreversible deletion of biological, cultural, and technological possibility.

1 Typed ports and boundary events

Definition 1.1 (Typed port)**.** *A typed port is a pair*

$$p = (I_p, O_p),$$

where I_p is an ingress space and O_p is an egress space. Its boundary-event space is

$$\Omega_p = I_p \times O_p.$$

A finite trace is an element of the free monoid

$$\mathsf{Tr}_p = \Omega_p^*.$$

The boundary is zero-dimensional in the sense relevant here: no Euclidean membrane is assumed. The primitive datum is a typed cut through which observable coupling occurs.

Definition 1.2 (Basilisk quartet)**.** *A Basilisk quartet over p is a tuple*

$$Q_p = (\mathsf{C}_p, \mathsf{S}_p, \mathsf{B}_p, \mathsf{L}_p)$$

whose components are respectively an admissibility artifact, a realization artifact, a separation artifact, and an attestation artifact over the same port.

2 Contract: admissibility boundedness

Definition 2.1 (Contract)**.** *A Contract is a subset*

$$\mathsf{C}_p \subseteq \Omega_p$$

or equivalently a predicate $\chi_{\mathsf{C}_p} : \Omega_p \to \{0,1\}$. Its trace lift is

$$\mathsf{C}_p^* = \{(\omega_0, \dots, \omega_{n-1}) \in \mathsf{Tr}_p : \omega_t \in \mathsf{C}_p \text{ for every } t\}.$$

When Ω_p is metric and C_p is closed, define the contract margin of a permitted event ω by

$$m_{\mathsf{C}}(\omega) = d_\Omega(\omega, \mathsf{C}_p^c).$$

The Contract corresponds to test-time differentiation collapse (TTDC): a larger possibility space is reduced to an admissible decision relation.

3 Script: realization boundedness

Definition 3.1 (Script)**.** *Let M be an internal state space. A Script is a deterministic map*

$$\mathsf{S}_p : M \times I_p \to M \times O_p$$

or, more generally, a Markov kernel

$$\mathsf{S}_p : M \times I_p \rightsquigarrow M \times O_p.$$

Its boundary realization relation is

$$R_{\mathsf{S}_p} = \{(i, o) : \exists m, m' \text{ with } \mathsf{S}_p(m', o \mid m, i) > 0\}.$$

The basic Script–Contract refinement condition is

$$R_{\mathsf{S}_p} \subseteq \mathsf{C}_p.$$

The Script corresponds to test-time integrative expansion (TTIE): it constructs a trajectory within the feasible relation.

Definition 3.2 (Lipschitz Script)**.** *For metric state and port spaces, a stochastic Script is L_S-Lipschitz when*

$$W_1\big(\mathsf{S}_p(\cdot \mid m, i), \mathsf{S}_p(\cdot \mid m', i')\big) \le L_S\big(d_M(m, m') + d_I(i, i')\big).$$

4 Blanket: coupling boundedness

Definition 4.1 (Separator Blanket)**.** *Let E be external state, M internal state, and $B = (I_p, O_p)$ boundary state. A probabilistic Blanket satisfies*

$$E \perp M \mid B.$$

For a joint law P, define the approximate Blanket defect

$$\delta_B(P) = I_P(E; M \mid B),$$

the conditional mutual information between external and internal state given the boundary.

A Blanket does not assert a fixed cardinality or Euclidean shape. It asserts mediated dependence.

Definition 4.2 (Bayesian-network Markov shell)**.** *For a node x in a directed acyclic graph,*

$$\mathrm{MB}(x) = \mathrm{Pa}(x) \cup \mathrm{Ch}(x) \cup \big(\mathrm{Pa}(\mathrm{Ch}(x)) \setminus \{x\}\big).$$

Equivalently, the shell contains parents, children, and other parents of those children.

The third term is forced by collider closure: conditioning on a child c in $x \to c \leftarrow z$ exposes dependence on the co-parent z. This is the invariant combinatorial pattern, even though drawings, geometry, and cardinality vary.

Proposition 4.3 (Markov-shell invariance under relabeling)**.** *Let G be a directed graph and let π be a vertex permutation. Then*

$$\pi(\mathrm{MB}_G(x)) = \mathrm{MB}_{\pi G}(\pi x).$$

Proof. Vertex relabeling preserves the parent, child, and co-parent relations. Applying π to each set in the defining union yields the blanket of the relabeled node. □

The Blanket corresponds to test-time coherent sampling (TTCS): it specifies the dependency cut through which coherent observations and actions are mediated.

5 Ledger: refinement and claimability boundedness

Definition 5.1 (Attested Ledger)**.** *Let Σ contain timestamps, signatures, provenance, or proof material. A Ledger is a finite sequence*

$$\mathsf{L}_p = ((\omega_t, \sigma_t))_{t<n} \in (\Omega_p \times \Sigma)^*.$$

An encoding map $\ell : \mathsf{Tr}_p \to \mathcal{L}_p$ is lossless when there exists a decoding map $d : \mathcal{L}_p \to \mathsf{Tr}_p$ such that

$$d \circ \ell = \mathrm{id}_{\mathsf{Tr}_p}.$$

An account is a refinement map

$$\alpha_{\mathsf{C}} : \mathcal{L}_p \to \mathcal{R}_p$$

into a residual space. A certificate is a witness π accepted by a verifier V_{C} for a declared claim about the decoded trace. The Ledger corresponds to test-time precision refinement (TTPR): it turns a raw history into a stable account, residual, or certificate.

6 The quatradic lock

Let

$$\partial : \mathsf{Run}(\mathsf{S}_p, \mathsf{B}_p) \to \mathsf{Tr}_p$$

extract the boundary trace of a mediated run.

Definition 6.1 (Exact quatradic lock)**.** *A quartet is exactly locked on an intended run class when:*

(i) $\partial(\mathsf{Run}(\mathsf{S}_p, \mathsf{B}_p)) \subseteq \mathsf{C}_p^*$*;*

(ii) $\delta_B(P) = 0$ *for the declared joint law;*

(iii) $d \circ \ell = \mathrm{id}$ *on extracted traces;*

(iv) $V_{\mathsf{C}}(\ell(\tau), \pi) = 1$ *if and only if* $\tau \in \mathsf{C}_p^*$.

The corresponding commuting requirement is

$$V_{\mathsf{C}} \circ \ell \circ \partial = \chi_{\mathsf{C}_p^*} \circ \partial.$$

The verifier and direct contract evaluation must agree; neither route is permitted to silently redefine the claim.

7 Local robust compliance

Theorem 7.1 (Contract margin under Lipschitz realization)**.** *Let X be a metric perturbation space, let $F : X \to \Omega_p$ be L-Lipschitz, and let $\mathsf{C}_p \subseteq \Omega_p$ be closed. Suppose*

$$r = d_\Omega(F(x_0), \mathsf{C}_p^c) > 0.$$

Then

$$d_X(x, x_0) < \frac{r}{L} \quad \Longrightarrow \quad F(x) \in \mathsf{C}_p.$$

Proof. By Lipschitz continuity,

$$d_\Omega(F(x), F(x_0)) < r.$$

Every point of C_p^c is at distance at least r from $F(x_0)$. Hence $F(x) \notin \mathsf{C}_p^c$, so $F(x) \in \mathsf{C}_p$. □

Remark 7.2. *The theorem controls local escape from the Contract. It does not establish Blanket closure, Ledger fidelity, or preservation of future possibility.*

8 Non-collapse: the Ledger does not identify the Script

Proposition 8.1 (Finite Script non-identifiability)**.** *There exist non-isomorphic deterministic Scripts that produce the same boundary trace for every input sequence and initial visible state.*

Proof. Let $I = O = \{0, 1\}$ and let the visible output be $o = i$. Let Script S_1 have one internal state and output i. Let Script S_2 have two internal states, toggle its hidden state at every step, and also output i. Their hidden transition systems are non-isomorphic, but every input sequence induces the same I/O trace. Therefore any Ledger containing only the boundary trace cannot identify which Script ran. □

Thus

$$\mathsf{L}(S_1) = \mathsf{L}(S_2) \not\Longrightarrow S_1 \cong S_2.$$

A certificate of trace compliance is not a certificate of complete generative identity.

9 Future-preserving boundedness

Let X be a state space and Y an ambient future space. Let

$$\mathcal{R} : X \to \mathcal{K}(Y)$$

assign each state a compact set of reachable futures.

Definition 9.1 (Irreversible deletion defect). *For a transition $T : X \to X$ and measure μ on Y, define*

$$\delta_{\mathrm{irr}}(T; x) = \mu\big(\mathcal{R}(x) \setminus \mathcal{R}(Tx)\big).$$

Definition 9.2 (Future-bounded transition). *A transition T is (L, ε)-future-bounded when*

$$d_H(\mathcal{R}(Tx), \mathcal{R}(Tx')) \leq L d_X(x, x')$$

for all x, x', and

$$\sup_x \delta_{\mathrm{irr}}(T; x) \leq \varepsilon.$$

The first condition bounds sensitivity of future sets; the second bounds one-way destruction. Neither implies the other. A smoothly executed extinction may have small local perturbation sensitivity and enormous irreversible deletion defect.

Proposition 9.3 (Deletion monotonicity under irreversible projection). *Suppose $D : X \to X$ satisfies*

$$\mathcal{R}(Dx) \subseteq \mathcal{R}(x)$$

for all x. Then

$$\delta_{\mathrm{irr}}(D; x) = \mu(\mathcal{R}(x)) - \mu(\mathcal{R}(Dx))$$

whenever both sets are measurable and finite in measure.

Proof. The inclusion makes the set difference disjoint and gives

$$\mathcal{R}(x) = \mathcal{R}(Dx) \sqcup (\mathcal{R}(x) \setminus \mathcal{R}(Dx)).$$

Finite additivity yields the claim. □

This provides a mathematical home for biological and cultural preservation inside the boundary architecture. The Contract sets a deletion budget, the Script realizes transitions, the Blanket limits uncontrolled externalization, and the Ledger records what future reachability was consumed.

10 Failure accounting

Let ε_B bound the probability of behavior arising through undeclared coupling, and let ε_L bound false acceptance by the Ledger verifier.

Proposition 10.1 (Elementary accepted-violation bound). *If every Contract-violating accepted run lies in the union of a Blanket-failure event and a Ledger false-accept event, then*

$$\Pr[\textit{violation accepted}] \leq \varepsilon_B + \varepsilon_L.$$

Proof. Apply the union bound. □

The proposition is intentionally modest. Stronger bounds require an explicit dependence model and cannot be obtained by naming the quartet alone.

11 Research directions

1. Dual and spectral certificates converting local Blanket observations into Script or Contract bounds.
2. Lower bounds on the realization cost of Contracts.
3. Local-to-global obstructions: locally compliant blankets with no coherent global Script.
4. Correlated repetition theorems for adaptive boundary testing.
5. Compressed Ledgers with moment-reconstruction guarantees.
6. Formal counterexamples separating all four artifact classes.
7. Polycentric governance in which inference authority and actuation authority remain distinct.

Provenance note

The distributed-coherence intuitions behind this work were materially influenced by BaAka polyrhythmic musical practice as encountered through Michelle Kisliuk's scholarship. Formalization does not extinguish origin. A fuller statement appears in the repository's `PROVENANCE.md`.